Hospital Dashboard Analysis Report

Data insights and operational recommendations for Jan 1 – Dec 20, 2019.

Speaker Notes: Welcome to the 2019 hospital dashboard analysis. This deck synthesizes our visit and mortality data to provide clear, actionable operational steps for management.

Three critical takeaways shape our operational strategy



The facility managed **403 visits** with a **17.1%** overall mortality rate.



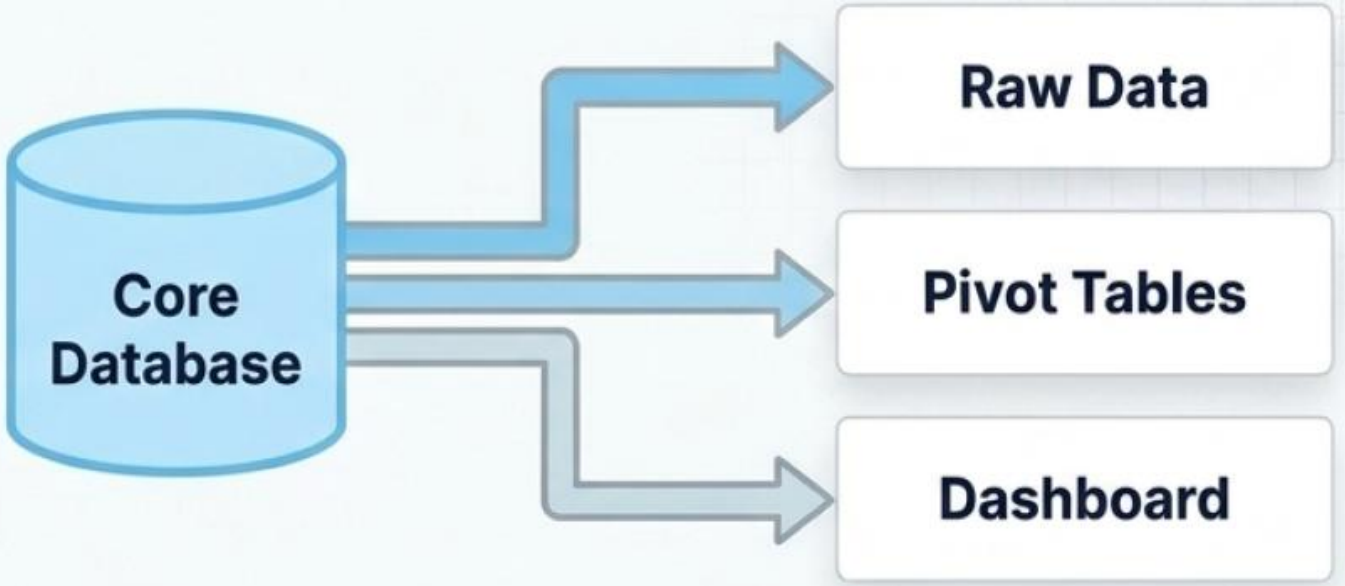
August recorded a peak 29% mortality, and **high risk** shows borderline association.



Shift to quarterly reporting and immediately **collect richer diagnostic data**.

Speaker Notes: If you take away only three things from this report, let it be these. We established our baseline, identified an anomaly in August, and formulated clear steps for data gathering.

The analysis is built on a clean, missing-value-free dataset



- Dataset spans 403 unique visits across 3 distinct structural sheets.
- Data is strictly clean, containing zero missing values or duplicate IDs.
- Minor formatting artifacts are present but do not impact integrity.

Speaker Notes: Before diving into insights, we must validate our source. The dataset is exceptionally clean and structured across three sheets, giving us high confidence in the baseline calculations.

2019 Baseline: 403 visits with a 17.1% mortality rate

403
Total Visits

69
Total Deaths

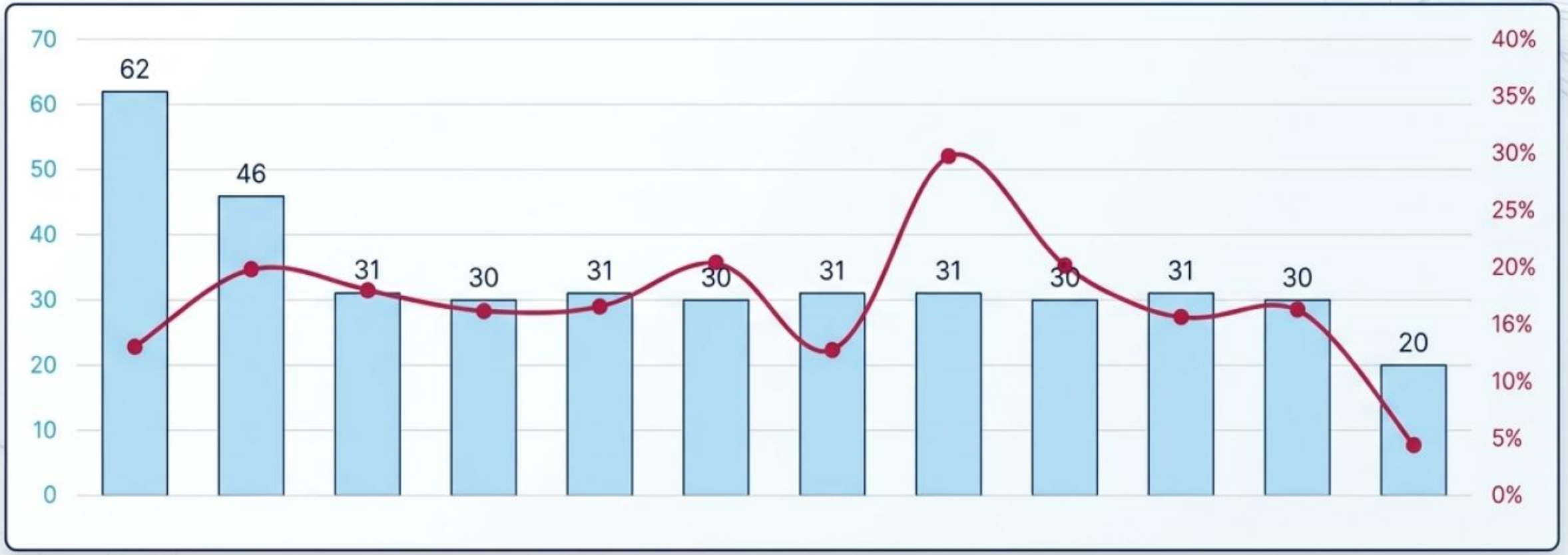
17.1%
Mortality Rate

45.2%
High-Risk Visits (182)

Speaker Notes: Here is our macro view for the year. We processed 403 visits. Nearly half were classified as high-risk, and our overall mortality rate sits at 17.1%.

Visit volumes peaked in January before stabilizing to ~30 per month

- January recorded the maximum volume with 62 visits.
- From March to November, volumes stabilized to approximately 30 per month.
- Visit distribution remains remarkably even across all days of the week.



Speaker Notes: Looking at throughput, we experienced a significant surge in Q1, particularly January. For the remainder of the year, our operational load stabilized.

August recorded the highest mortality rate of the year at 29%

- August yielded 9 deaths out of just 31 visits, establishing the yearly peak.
- February matched the 9 absolute deaths, but on a larger volume of 46 visits.
- January and July recorded the lowest mortality rates at 12.9%.



Speaker Notes: August requires attention. While February also saw 9 deaths, August saw the same absolute number against a much smaller patient volume.

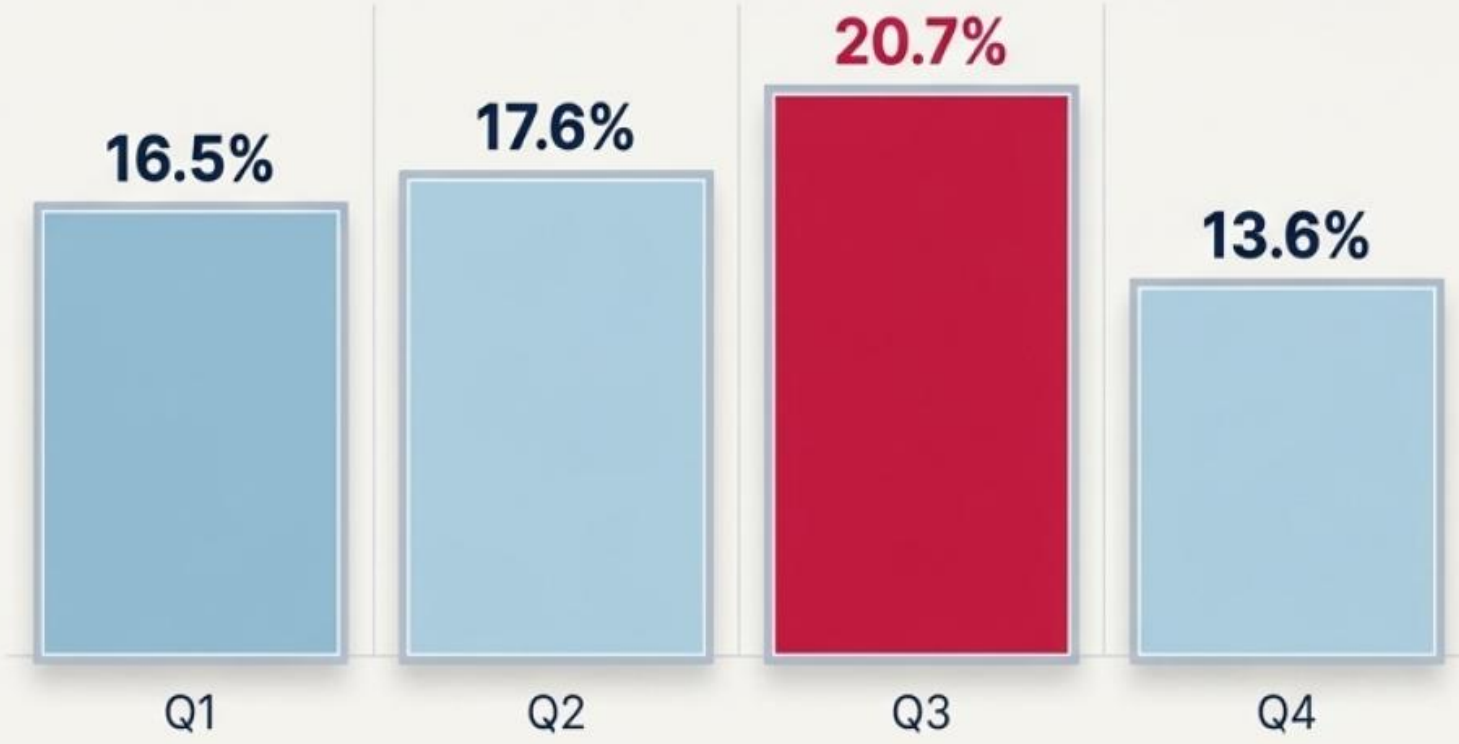
High-risk classification is associated with higher mortality, but is borderline



- High-risk visits experienced a 21.4% mortality rate versus 13.6% for low-risk.
- This represents an approximately **1.6x multiplier** between the categories.
- The difference is statistically borderline and does **not prove** definitive causation.

Speaker Notes: While high-risk patients show higher mortality, the p-value means this association is borderline and does not prove causation on its own.

A quarterly view smooths monthly noise and highlights a Q3 peak



- Q1 and Q2 maintained relatively stable mortality rates at 16.5% and 17.6%.
- Q3 experienced the highest quarterly rate at **20.7%**, driven by the **August peak**.
- Q4 recorded the lowest rate at 13.6%, skewed by incomplete December data.

Speaker Notes: Zooming out to a quarterly view provides a truer trend. Q3 was our most challenging period, while Q4 appears lowest, though incomplete.

Small sample sizes and incomplete data limit definitive conclusions

December data terminates on the 20th, preventing fair comparison with other months.

Small monthly denominators mean a single death arbitrarily swings rates by ~3%.

Gaps in visit IDs indicate this dataset is likely a sample, not the entire hospital.

Speaker Notes: We must acknowledge the boundaries of our data. Small sample sizes mean a single outcome drastically swings percentages, and December is truncated.

Shift to quarterly reporting and expand diagnostic data collection

- Adopt quarterly reporting to mitigate the statistical noise of small monthly samples.
- Initiate a targeted operational review of August cases to identify systemic drivers.
- Enrich future datasets with patient age, department, diagnosis, and length of stay.

Speaker Notes: To move forward, we must stop reacting to monthly noise and shift to quarterly reporting. Furthermore, an immediate review of August is required.